

Supplementary Material: Cross-Track Sentinel-1 Land-Cover Classification with Two-Acquisition Interferometric Coherence

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TABLE S1

RF CONFIGURATIONS SELECTED WITH 400 M SPATIAL EXCLUSION.

Repr.	Track	m_{features}	Min. leaf	CV BA (%)
I_8	T15	$\sqrt{D}$	1	68.06
I_8	T37	$\sqrt{D}$	1	68.84
I_8	T88	0.5	1	67.48
I_8	T139	0.5	1	69.35
I_{28}	T15	0.5	1	70.74
I_{28}	T37	$\sqrt{D}$	1	71.85
I_{28}	T88	0.5	1	69.99
I_{28}	T139	$\sqrt{D}$	1	72.95
$I_8 + g_{12}$	T15	$\sqrt{D}$	1	77.52
$I_8 + g_{12}$	T37	0.5	1	76.58
$I_8 + g_{12}$	T88	0.5	1	75.81
$I_8 + g_{12}$	T139	$\sqrt{D}$	1	77.32
F_{12}	T15	0.5	1	77.81
F_{12}	T37	$\sqrt{D}$	1	77.07
F_{12}	T88	$\sqrt{D}$	1	74.13
F_{12}	T139	0.5	5	77.47
$I_{28} + g_{12}$	T15	0.5	1	78.08
$I_{28} + g_{12}$	T37	0.5	1	77.86
$I_{28} + g_{12}$	T88	$\sqrt{D}$	1	75.99
$I_{28} + g_{12}$	T139	$\sqrt{D}$	1	78.39

S-I. SOURCE-TRACK MODEL SELECTION

Tables S1 and S2 list the 40 configurations selected by buffered source-track spatial cross-validation. Each configuration is then fitted with training-sample seeds 17, 29, and 43. The RF uses 300 trees; the SVC uses an RBF kernel and a source-fitted robust scaler. SVC entries report the multiplier of γ_{scale} . The displayed validation accuracy is the mean across the three source spatial folds.

S-II. LABEL AND SUPPORT SENSITIVITY

The label sensitivity experiment uses the primary training and evaluation blocks with a separate sample on U_2 , the support with valid labels in both products. Up to 800 pixels per evaluation block are drawn with seed $42 + b$, where b is the block identifier. Restricting this sample to product agreement defines its C_2 subset. Training candidates within 400 m of any sampled evaluation pixel are excluded; each final fit uses 20,000 pixels and the frozen source-track hyperparameters. Thus, the C_2 sensitivity sample differs from the primary 9,600-pixel consensus sample.

WorldCover (WC), CORINE (CLC), and consensus labels are combined into six training-evaluation designs. Table S3 reports mean paired increments over twelve directed track pairs

TABLE S2

SVC CONFIGURATIONS SELECTED WITH 400 M SPATIAL EXCLUSION.

Repr.	Track	C	γ multiplier	CV BA (%)
I_8	T15	100.0	0.1	66.95
I_8	T37	10.0	0.1	68.12
I_8	T88	1.0	1.0	66.28
I_8	T139	100.0	0.1	68.83
I_{28}	T15	1.0	0.1	70.44
I_{28}	T37	10.0	0.1	71.04
I_{28}	T88	10.0	0.1	69.46
I_{28}	T139	100.0	0.1	72.12
$I_8 + g_{12}$	T15	10.0	0.1	79.73
$I_8 + g_{12}$	T37	10.0	0.1	77.38
$I_8 + g_{12}$	T88	10.0	0.1	76.08
$I_8 + g_{12}$	T139	10.0	0.1	77.13
F_{12}	T15	10.0	0.1	79.32
F_{12}	T37	10.0	0.1	77.44
F_{12}	T88	100.0	0.1	75.61
F_{12}	T139	10.0	0.1	77.07
$I_{28} + g_{12}$	T15	10.0	0.1	80.33
$I_{28} + g_{12}$	T37	10.0	0.1	77.90
$I_{28} + g_{12}$	T88	10.0	0.1	77.91
$I_{28} + g_{12}$	T139	100.0	0.1	78.19

and three sampling seeds. Bracketed values are the empirical 2.5th and 97.5th percentiles of these 36 deployment increments. They describe variation among the evaluated deployments and are not confidence intervals for a population mean. Primary spatial block-bootstrap intervals are reported separately in the main article.

S-III. SPATIAL RESAMPLING AND TEMPORAL EVALUATION

Primary spring contrasts use a paired bootstrap of twelve evaluation blocks with replacement. Each draw uses the same block multiplicities for the compared representations, tracks, and seeds. Of 20,000 draws, 19,306 contain all five classes; spring BA intervals use this subset. The other 694 draws omit water. The evaluation sample contains 1,935 forest, 5,302 grassland, 2,140 cropland, 165 built-up, and 58 water pixels.

The autumn analysis uses the frozen spring RF and SVC models for I_{28} and $I_{28} + g_{12}$, yielding 48 fitted model configurations and 1,843,200 prediction rows across all targets. No autumn labels enter model fitting or selection. Cross-track metrics exclude each model's source-track target. The prediction table contains 9,600 unique evaluation pixels.

The autumn bootstrap retains all 20,000 block draws. In the 633 draws with an absent class, BA averages recall over classes present in both compared arms. BA intervals are bias-corrected and accelerated (BCa), using leave-one-block-out estimates for

TABLE S3
BUFFERED LABEL/SUPPORT SENSITIVITY: MEAN Δ BA AND CENTRAL 95% DEPLOYMENT RANGE (PP).

Training label	Evaluation label	Support	RF	SVC
WC	WC	U2	+3.64 [0.68, 6.79]	+6.68 [3.95, 10.02]
WC	CLC	U2	+5.54 [2.97, 7.92]	+8.76 [5.04, 12.61]
CLC	CLC	U2	+6.53 [4.61, 9.56]	+8.46 [5.59, 13.67]
CLC	WC	U2	+4.14 [2.28, 6.73]	+5.69 [2.61, 10.09]
WC	WC	C2	+6.96 [4.37, 10.66]	+11.07 [6.73, 15.96]
C2	C2	C2	+7.02 [4.21, 10.45]	+7.60 [3.79, 11.07]

TABLE S4
AUTUMN PAIRED INCREMENTS AND 95% INTERVALS (PP).

Model	Metric	Increment	Interval
RF	BA	5.03	[3.33, 7.08]
SVC	BA	5.23	[3.36, 7.27]
RF	OA	0.62	[-0.36, 1.74]
SVC	OA	-0.01	[-0.68, 0.79]

deployment records under

`tgrs_revision/results/robustness/`.

- Autumn predictions: `external_validation/results/prediction_long.parquet`; statistics: `corrected_bootstrap.csv` in the same directory.

acceleration; OA intervals are percentile intervals. Table S4 reports full-sample point estimates separately from bootstrap summaries.

S-IV. RADIOMETRIC PROCESSING CONVENTION

RTC products are converted to a common dB representation before interpolation to the analysis grid. T15 and T37 products are delivered in dB; T88 and T139 products are delivered in linear power. The latter undergo $10\log_{10}$ conversion at native resolution before bilinear interpolation to 40 m. This order applies the same radiometric operation across tracks. Conversion after interpolation would yield a different result because logarithms and spatial averaging do not commute. The processing implementation is retained in `sar_v2/data/build_4track_cube_LOCAL_v3.py`.

S-V. GEOGRAPHIC REPLICATION FEASIBILITY

The campaign records two Friesland candidate boxes: Box A spans 5.30–5.85°E and 52.80–53.20°N; Box B spans 5.30–5.84°E and 53.00–53.40°N. The archive-query record reports that T139 does not cover these candidates. Reproducing the Site-1 configuration would therefore require a different orbit set. The geographic evaluation remains unexecuted; the reported external acquisition experiment is the same-site autumn evaluation. The candidate definitions and coverage issue are recorded in `external_validation/NEEDS_AUTHOR_DECISION.md`.

S-VI. ARTIFACT MAP

Paths below are relative to the project root. The primary campaign directory is `tgrs_final_campaign_20260915`.

- Main models and pixel predictions: `campaign_runs_R1/`.
- Selected parameters: `campaign_configs/R1_selected_models.json`.
- Primary contrasts: `campaign_tables/T1_contrasts.csv`; per-class results: `tables/T1_per_class.csv`.
- Label sensitivity: `tgrs_revision/tables/table_label_sensitivity_buffered.csv`, with

S-VII. TRACK AND SPATIAL DIAGNOSTICS

The complete transfer matrix and the following figures retain the pair-level and spatial diagnostics accompanying the main results. They show angular dependence, transfer-gap contrasts, a second spatial example, object concentration, and block-level increments.

TABLE S5

FULL 4×4 CROSS-TRACK TRANSFER MATRIX IN BALANCED ACCURACY (%) UNDER THE 400 M-BUFFERED PROTOCOL, POOLED OVER THREE SAMPLING SEEDS. EACH CELL READS $I_{28} \rightarrow I_{28} + g_{12}$, WITH THE INCREMENT IN PARENTHESES. DIAGONAL (BOLD) CELLS ARE IN-TRACK.

Source	Random Forest $\rightarrow$ target track				SVC $\rightarrow$ target track			
	T15	T37	T88	T139	T15	T37	T88	T139
T15	72.0 $\rightarrow$ 78.8 (+6.8)	66.5 $\rightarrow$ 73.1 (+6.6)	70.3 $\rightarrow$ 77.2 (+6.9)	66.9 $\rightarrow$ 73.1 (+6.2)	71.0 $\rightarrow$ 80.2 (+9.2)	66.4 $\rightarrow$ 74.6 (+8.2)	69.5 $\rightarrow$ 77.1 (+7.6)	67.3 $\rightarrow$ 73.3 (+6.0)
T37	70.8 $\rightarrow$ 77.8 (+7.0)	70.2 $\rightarrow$ 76.0 (+5.8)	70.4 $\rightarrow$ 76.5 (+6.1)	68.8 $\rightarrow$ 73.5 (+4.7)	70.8 $\rightarrow$ 78.2 (+7.4)	70.6 $\rightarrow$ 76.5 (+6.0)	70.9 $\rightarrow$ 78.0 (+7.2)	68.1 $\rightarrow$ 74.0 (+6.0)
T88	67.7 $\rightarrow$ 74.6 (+6.9)	65.1 $\rightarrow$ 71.2 (+6.1)	71.8 $\rightarrow$ 77.1 (+5.3)	66.8 $\rightarrow$ 72.5 (+5.7)	67.8 $\rightarrow$ 75.4 (+7.6)	66.0 $\rightarrow$ 73.2 (+7.2)	71.6 $\rightarrow$ 78.1 (+6.5)	65.5 $\rightarrow$ 72.0 (+6.6)
T139	65.7 $\rightarrow$ 72.0 (+6.3)	60.9 $\rightarrow$ 68.2 (+7.4)	64.5 $\rightarrow$ 70.3 (+5.8)	73.0 $\rightarrow$ 77.8 (+4.9)	65.8 $\rightarrow$ 72.2 (+6.4)	59.7 $\rightarrow$ 67.3 (+7.6)	62.8 $\rightarrow$ 69.1 (+6.3)	72.7 $\rightarrow$ 77.8 (+5.1)
Mean	cross-track 67.03 $\rightarrow$ 73.33 (+6.30); in-track 71.78 $\rightarrow$ 77.44 (+5.66)				cross-track 66.72 $\rightarrow$ 73.71 (+6.99); in-track 71.46 $\rightarrow$ 78.15 (+6.69)			

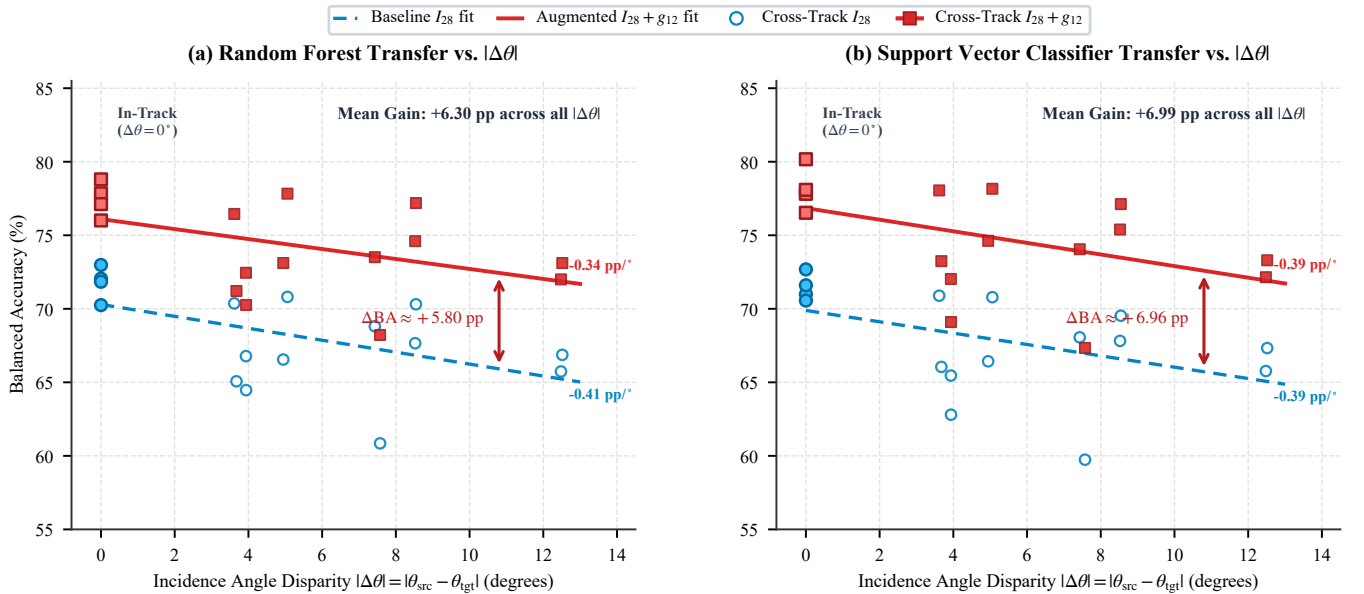


Fig. S1. Balanced accuracy against source–target incidence-angle separation for (a) RF and (b) SVC across sixteen track pairs. Baseline and augmented regression slopes are -0.41 and -0.34 $\text{pp}/^\circ$ for RF, and -0.39 and -0.39 $\text{pp}/^\circ$ for SVC. Lines summarize the observed track configurations.

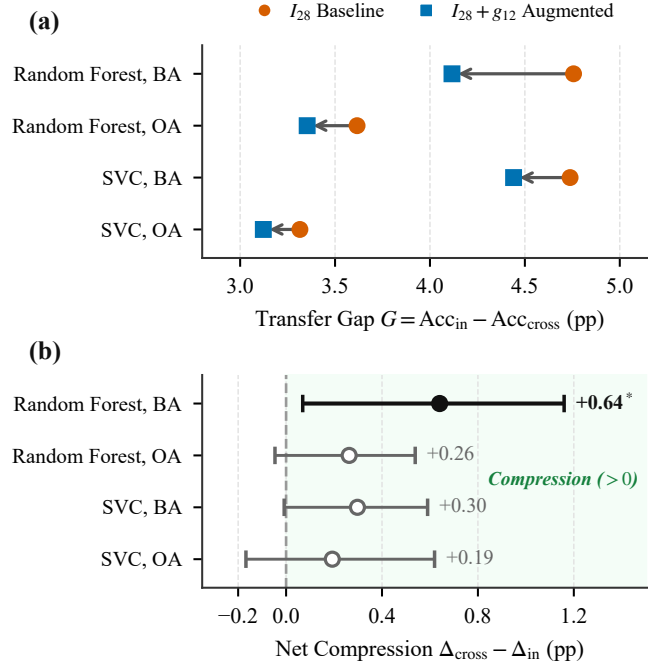


Fig. S2. (a) In-track minus cross-track accuracy before and after coherence augmentation. (b) Paired gap reductions with 95% joint block bootstrap intervals. The RF balanced-accuracy reduction is +0.64 pp [+0.07, +1.16]; the other three intervals include zero.

Evaluation Block 803 (Boekelo Suburban Fringe — Peak Sign-Test Gain $\Delta\text{BA} = +16.1$ pp, Built-Up Recall Gain +24.2 pp)

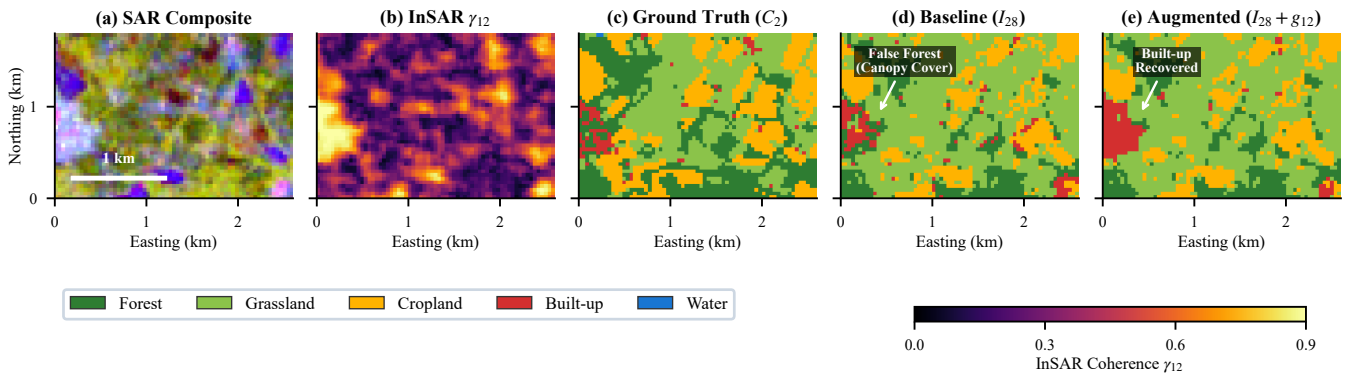


Fig. S3. Spatial error resolution in suburban landscape under cross-track transfer (T15 → T37, steep ascending to medium descending) in Block 803 (Boekelo suburban fringe, peak sign-test gain $\Delta\text{BA} = +16.1$ pp): (a) SAR false-colour composite, (b) 12-day InSAR coherence γ_{12} , (c) harmonized consensus ground truth C_2 , (d) baseline I_{28} predictions misclassifying residential housing under canopy as forest (red outline), and (e) coherence-augmented $I_{28} + g_{12}$ predictions restoring the suburban core (+24.2 pp local built-up recall gain; rural Block 603 exhibits an analogous +37.5 pp gain).

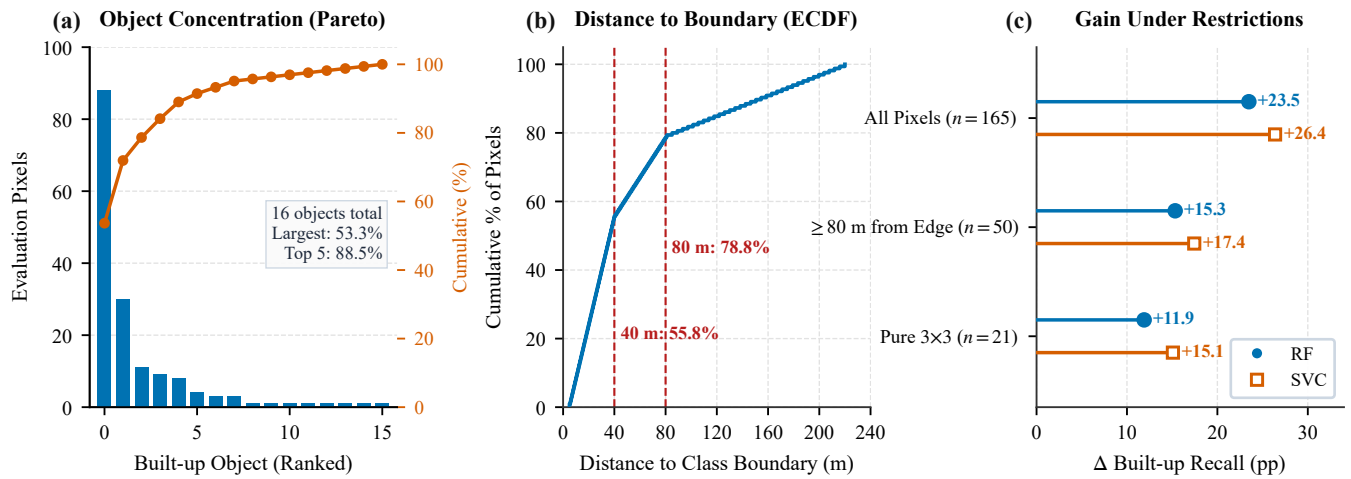


Fig. S4. Spatial support of the 165 built-up evaluation pixels. (a) Contributions of 16 connected objects; the largest supplies 53.3% and the five largest 88.5%. (b) Distance to class boundaries. (c) Recall increments on interior and locally homogeneous subsets.

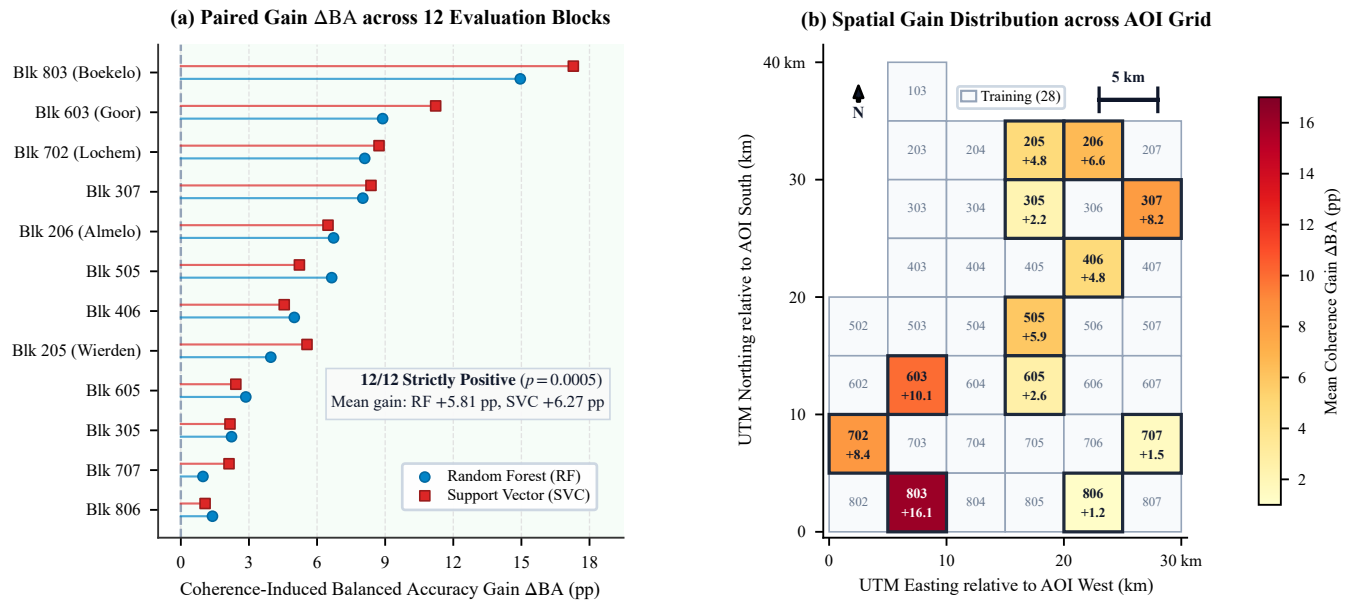


Fig. S5. Block-level coherence increments across twelve evaluation blocks. (a) RF and SVC increments, positive in all blocks for each learner (two-sided sign test, $p = 0.0005$). (b) Spatial distribution of the mean increment, with training blocks shown for context.